Facing, Composition and the Center in Native Binary Rings

A technical manual for reading interfaces in both directions

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Abstract

Native binary rings list every word of LEN n once and place each word opposite its facing partner. The facing law reads a word backward and, where that reading changes nothing, takes its NOT: $P_n(w) = J_n(w) \oplus s_n(w) \cdot 1^n$, with J the reversal and s the selector of words that read the same both ways. This manual defines the objects, gives the procedures, and states the results with proofs and exhaustive checks.

For a contact that reads differently backward, facing exchanges interface and implementation: the partner receives what the contact hands on and hands on what it receives, each read backward. NOT supplies the partner on a share $2^{\lceil n/2 \rceil - n}$ of contacts, all of LEN 1 and almost none at large LEN. Facing is a pure NOT at LEN 1 and 2, affine at LEN 3, and of algebraic degree $\lfloor n/2 \rfloor$ beyond. A contact composed with its own backward reading, through a hinge that reads the same both ways, forms a whole that reads the same both ways, so the whole is faced by NOT. Every non-constant palindrome one level up arises this way exactly once, and in both contact models the facing of a composed whole does not factor through its parts. Facing preserves the number of contacts and the offset between LENs doubles it, so the elementary change of count is one bit.

The center of a word is a free bit at odd LEN and a seam at even LEN. At the center of the ring, the edge across the seam joins a word to its own reversal and is faced by NOT. A register that pairs each contact with its backward reading reads out exactly along Page's curve. The one-bit step, sealed as a prediction for black-hole horizons and checked against the GW250114 ringdown, survives; the 8π step fails in the emission-line model used; binary against ternary is open.

Contents

- 1 The founding claim and how to use this manual
- 2 How the result was reached

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- 3 Objects and notation
- 4 Procedures
- 5 Facing is the other reading
- 6 Composition
- 7 Count, offset and closure
- 8 The center
- 9 The paired register and its readout curve
- 10 Landing: the NOT at the Planck floor and the horizon register
- 11 The interface catalog and the motion reader
- 12 Technical notes
- 13 Ledger
- 14 How it continues
- Appendix A Seals
- Appendix B Reference implementation
- Appendix C Reproduction
- References

1 The founding claim and how to use this manual

This manual holds one claim fixed. Difference is existence, and the first distinction is NOT. NOT forces binary: every base is a reading of binary, and the values a process can produce depend on where in the ring the process sits. Chained difference routes, that is transformations, are the constitutive substrate of reality. They are not a method applied to a domain that exists without them.

An interface is a need: the contract for what must happen across a contact. An implementation reflects its interface by what it does — where it moves, how much space it uses, when it comes back — and not by the values it emits; values are transfer objects between layers. Parameters are typed by LEN, not by kind. The native rings list every way to read difference in both directions, offset by one. The interface at one LEN is the implementation at the next, and the direction of reading tells interface and implementation apart.

The manual is organized like software documentation. Section 3 defines the types. Section 4 gives the procedures, each with input, steps, output and an example. Sections 5 to 11 state the results with proofs and exhaustive checks. Section 12 holds the attributed technical notes, Section 13 the ledger and Section 14 the next operations. Appendix A gives the seals, Appendix B the reference implementation and Appendix C the reproduction steps.

Each result carries one status label. **STANDS**: proved for every LEN. **RUNS**: checked on every word in the stated range. **measured**: read from outside data. **ELSEWHERE**: the result belongs to a different owner than first assigned. **OPEN**: named and not yet derived. **PULL**: a direction the program sets, with a defined next test.

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2 How the result was reached

Hold the finished structure in view: one ring, read in two directions, with NOT supplying the partner exactly where the two readings agree. Each piece was forced by the one before it, and the path back runs as follows.

The first distinction is NOT, and a distinction has to be paid for. A finer distinction costs more, and the payment bends the field that has to read it. The two requirements run in opposite directions and cross at one scale, the Planck floor (Section 10.1). A surface that holds as many distinctions as its area allows is a horizon, so the floor led to the horizon, and the horizon asked for its elementary step. An independent audit (Technical Note 1) separated two operations that the step had merged: flipping an existing distinction, which keeps the count, and adding an independent one, which doubles it. The ring already contained both. Facing acts inside one LEN and keeps the count; the offset acts between LENs and doubles it (Section 7).

Mapping facing on every contact showed that it is the other reading, with NOT standing in only where the two readings coincide (Section 5). Composing contacts with their own partners showed where those coincidences come from: they are built one level down (Section 6). The same audit derived a whole-versus-parts law in the §181 transition table; it is the composition law of the offset wheel at a different location (Section 6.3). The question of the center fixed the last piece. A word's center has no reflection at its own LEN, and its partner arrives with the next offset as a neighbor. At the center of the ring, the edge across the seam is its own reflection, so its other face is its difference (Section 8). Reading a register paired by reflection in order reproduces the curve expected from an evaporating horizon (Section 9), and the one-bit step was sealed and landed on a measured ringdown (Section 10).

3 Objects and notation

3.1 Words, positions and LEN

A word of LEN n is $w = w_0 w_1 \ldots w_{n-1} \in X_n = \{0,1\}^n$, written with position 0 on the left (the most significant side). Its value is $v(w) = \Sigma_i w_i 2^{n-1-i}$, and X_n has $N = 2^n$ words. The mirror of position i is n−1−i. At odd n exactly one position is its own mirror, the center $k = (n-1)/2$. At even n none is, and the center is the seam between positions n/2−1 and n/2. 1^n is the all-ones word and 0^n the all-zeros word.

3.2 Operators

Operators on X_n (as defined in Native Binary Rings and Bicomplex Difference Calculus on Native Binary Rings).

Symbol	Name	Definition	Properties
J_n	reversal (read backward)	$(J\,w)_i = w_{n-1-i}$	involution; linear
C_n	NOT	$C\,w = w \oplus 1^n$	involution; a translation
Pal_n	reads the same both ways	$\{w : J\,w = w\}$	subspace of dimension $\lceil n/2 \rceil$; $2^{\lceil n/2 \rceil}$ words
s_n	selector	$s_n(w) = \prod_{i<\lfloor n/2 \rfloor} (1 \oplus w_i \oplus w_{n-1-i})$	1 exactly on Pal_n; algebraic degree $\lfloor n/2 \rfloor$
P_n	facing (native partner)	$P_n(w) = J_n(w) \oplus s_n(w) \cdot 1^n$	fixed-point-free involution

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P_n reverses a word that reads differently backward and complements a word that reads the same both ways. A word whose partner is its reversal is **direction-faced**; a word whose partner is its NOT is **NOT-faced**. The selector is a product over mirror pairs only, so the center of an odd word never enters it.

3.3 Contacts in the offset model

A word e of LEN n+1 is a contact between two faces of LEN n: IN(e) = $e_0 \ldots e_{n-1}$, the interface it receives, and OUT(e) = $e_1 \ldots e_n$, the implementation it hands on. Contacts e and f chain when OUT(e) = IN(f); the chain is the word of LEN n+2 that reads e followed by the last bit of f. With X_n as faces and X_{n+1} as contacts this is the binary wheel of LEN n, the de Bruijn graph B(n): 2^n faces and 2^{n+1} contacts, each face with two contacts in and two out. Every contact of the wheel at LEN n is a face of the wheel at LEN n+1. This is the offset by one, under which the interface at one LEN is the implementation at the next.

3.4 Contacts in the juxtaposition model (§181)

In §181 a transition from face x to face y, both of LEN n, is the word x|y of LEN 2n. A path a→b→c is the word W = a|b|b|c of LEN 4n; it composes because its two middle faces are equal. Of the N^4 words of LEN 4n, exactly N^3 are composable paths. On the transition table, facing is P_{2n}: a transition that reads differently backward maps x|y to J(y)|J(x), and one that reads the same both ways maps x|y to C(x)|C(y).

3.5 The native ring

The 2^{n-1} facing pairs of X_n are ordered by their canonical representatives $r_0 < r_1 < \ldots < r_{m-1}$, where the representative of a pair is its smaller member and $m = 2^{n-1}$. The native ring R_n places them as follows (Native Binary Rings, eqs. 2–4):

$$R_n[a] = r_a \ (0 \le a < m-1), \quad R_n[m-1] = P_n(r_{m-1}), \quad R_n[m] = r_{m-1}, \quad R_n[a] = P_n(r_{N-1-a}) \ (m < a < N)$$

The address law $R_n[N-1-a] = P_n(R_n[a])$ holds at every address: the opposite address holds the other face. Addresses m−1 and m form the equatorial seam. For example R_2 = 00 | 10 | 01 | 11 and R_3 = 000 | 001 | 010 | 110 | 011 | 101 | 100 | 111. The native successor T_n advances the address by one. The twisted clock $\tau_n(x_0 \ldots x_{n-1}) = x_1 \ldots x_{n-1} \neg x_0$ (One Inverter) satisfies $\tau_n{}^n = C_n$ and $\tau_n{}^{2n} = I$.

3.6 Boundary and closure

Over F_2 the boundary of a contact is ∂e = IN(e) + OUT(e), a formal sum of faces. Along a chain the internal faces cancel, $\partial(e_1 + \ldots + e_k) = IN(e_1) + OUT(e_k)$, and a closed route has zero boundary. A connected structure with V faces and E contacts has E − V + 1 independent closed routes, its cycle rank.

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4 Procedures

Quick reference.

Task	Rule	Where
Face a contact	Read it backward; if nothing changed, take its NOT.	P1, §5
Read the exchange	Direction-faced: the partner has IN = J(OUT) and OUT = J(IN). NOT-faced: both faces negated.	P2, §5.1
Compose	Match faces: OUT(e) = IN(f) in the offset model; equal middle faces in §181.	P3, §6
Face a composed whole	Face it at its own LEN, never part by part.	P4, §6.3
Change the count	Only the offset changes it: one bit per LEN step.	§7
Find the center	Odd LEN: the middle bit. Even LEN: the seam. Ring: the seam between addresses m−1 and m.	P5, §8
Count closures	E − V + 1 for the connected structure.	P6, §7
Name an interface	Cycle type, order and fixed words decide its class in the catalog.	P7, §11.1
Identify an implementation	Compare motion (orbits, returns, places used), never values.	P8, §11.2
Seal a prediction	Write it, hash it, then look.	P9, App. A

P1 Face a contact. Input: a word w of LEN n. Steps: (1) read w backward to get J(w); (2) if J(w) ≠ w, the partner is J(w) and w is direction-faced; (3) otherwise the partner is C(w) and w is NOT-faced. Output: P(w) and the facing type. Example at LEN 3: 000↔111 (NOT), 001↔100 (direction), 010↔101 (NOT), 011↔110 (direction).

P2 Read the exchange. Input: a contact e of LEN n+1. Steps: face e with P1. If e is direction-faced, the partner's interface is J(OUT e) and its implementation is J(IN e). If e is NOT-faced, the partner's interface is C(IN e) and its implementation is C(OUT e). Output: the partner's two faces. Example: e = 001 runs 00 → 01; its partner 100 runs 10 → 00, that is J(01) → J(00).

P3 Compose two contacts. Offset model: e then f compose when OUT(e) = IN(f), and the whole is e followed by the last bit of f. Juxtaposition model: x|y then y|z compose into x|y|y|z. Example: 100 then 001 share the face 00 and compose into 1001.

P4 Face a composed whole. Face the whole at its own LEN with P1. Facing the parts and reassembling them gives a different result exactly in the cases counted in Section 6.3. Example (§181, n = 2): W = 00|11|11|00 reads the same both ways as a whole, so its partner is 11|00|00|11. Facing its parts 00|11 and 11|00 separately, each of which reads differently backward, returns 00|11|11|00, the original.

P5 Locate the center. Word of odd LEN n: position (n−1)/2, whose value is free among the palindromes and which the selector never reads. Word of even LEN: the seam between positions n/2−1 and n/2. Native ring: the seam between addresses m−1 and m, which holds J(r⋆) and r⋆ (Section 8.3).

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P6 Count closures. Count the faces V and contacts E of the connected structure; the independent closed routes number E – V + 1. A ring of k contacts has 1. The same contacts opened into a chain have 0. The wheel B(n) has $2^n + 1$.

P7 Name the interface of an affine transformation. Input: a map on X_L that respects XOR up to a translation. Steps: record its cycle type, its order and its number of fixed words, and find its conjugacy class in AGL(L,2). Output: its interface number in Table 7.

P8 Identify an implementation's interface by motion. Input: an implementation run from many starts. Steps: record orbit lengths, return lags read only as same or different, and the number of places each walk uses. Compare these between implementations; never compare histograms of the values emitted. Output: same interface or different (Section 11.2).

P9 Seal and land a prediction. Steps: (1) write the prediction, its derivation and its failure condition in a file; (2) compute the file's SHA-256 and record it with the UTC time; (3) only then open outside data; (4) label each part STANDS, RUNS, measured, ELSEWHERE or OPEN. Appendix A lists the seals used here.

5 Facing is the other reading

Theorem 5.1 (exchange). Let e be a contact of LEN n+1. If e is direction-faced, IN(P e) = J(OUT e) and OUT(P e) = J(IN e). If e is NOT-faced, IN(P e) = C(IN e) and OUT(P e) = C(OUT e).

Proof. Reversal sends position i to position n–i. The first n bits of J(e) are therefore the last n bits of e read backward, and the last n bits of J(e) are the first n bits of e read backward. Complement acts position by position. ■ **STANDS**; **RUNS** on every contact of LEN 2 to 14.

For a contact that reads differently backward, the partner receives what the contact hands on and hands on what it receives, each read backward. Interface and implementation are one set of contacts read in two directions. Where the two readings coincide, direction carries nothing and NOT supplies the partner.

Theorem 5.2 (the share of NOT). $|\mathrm{Pal}_n| = 2^{\lceil n/2 \rceil}$, so NOT faces $2^{\lceil n/2 \rceil - 1}$ of the 2^{n-1} pairs, a share of $2^{\lceil n/2 \rceil - n}$. **Proof.** A palindrome is fixed by its first $\lceil n/2 \rceil$ bits. ■ **STANDS**; **RUNS** LEN 1–14 (Table 1).

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Table 1. Facing pairs and the share faced by NOT.

LEN	Facing pairs	NOT-faced pairs	NOT share
1	1	1	1
2	2	1	1/2
3	4	2	1/2
4	8	2	1/4
5	16	4	1/4
6	32	4	1/8
7	64	8	1/8
8	128	8	1/16
9	256	16	1/16
10	512	16	1/32
11	1,024	32	1/32
12	2,048	32	1/64
13	4,096	64	1/64
14	8,192	64	1/128

The share forms a funnel. Reading direction fills the wide rings and NOT fills the center; at LEN 1 it is the only facing there is.

Theorem 5.3 (how nonlinear facing is). P_1 and P_2 are the translation by 1^n: at LEN 1 and 2 facing is a pure NOT. P_3 is affine and is not a translation; it is interface #3 of the LEN-3 catalog (Table 7). For $n \geq 2$ the selector s_n has algebraic degree $\lfloor n/2 \rfloor$, and from LEN 4 on so does P_n. The center of an odd word never enters s_n.

Proof. s_n is the product of $\lfloor n/2 \rfloor$ affine factors in disjoint pairs of variables, so its algebraic normal form contains the full product term. $P_n = J_n \oplus s_n \cdot 1^n$ adds only a linear map. The center appears in no factor. At LEN 2 the reversal of 01 is its NOT, so $P_2 = C_2$. ■ **STANDS**; **RUNS** LEN 1–12.

The test "reads the same both ways" is a conjunction over every mirror pair. That collective test is what makes facing nonlinear, and its degree grows with LEN.

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6 Composition

6.1 A contact composed with its own backward reading

Theorem 6.1. Let x be a direction-faced word of LEN L. Then x chains into J(x) if and only if OUT(x) reads the same both ways, and the chain then reads the same both ways. Likewise J(x) chains into x if and only if IN(x) reads the same both ways, and that chain also reads the same both ways. These ordered chains correspond one to one with the non-constant palindromes of LEN L+1, of which there are $2^{\lceil (L+1)/2 \rceil} - 2$.

Proof. IN(J x) = J(OUT x), so x chains into J(x) exactly when OUT(x) = J(OUT x). Write x = $x_0 \cdot p$ with p = OUT(x). The chain is x followed by the last bit of J(x), which is x_0, so the chain is $x_0 \cdot p \cdot x_0$, a palindrome when p is one. Conversely, every palindrome c of LEN L+1 has OUT(c) = J(IN c), so c is the chain of IN(c) into its own reversal. IN(c) reads the same both ways only when c is constant, and then IN(c) is NOT-faced, which excludes the two constant words. ■ **STANDS**; **RUNS** LEN 1–14 (Table 2).

A contact composed with its own other reading, through a hinge that reads the same both ways, is a whole that reads the same both ways, and its facing is therefore NOT. Direction, composed, turns into NOT one level up: the places where NOT faces a contact are built at the level below.

6.2 NOT pairs and the NOT clock

Theorem 6.2. A NOT-faced word x of LEN L chains into C(x) only when x alternates (0101… or 1010…), which requires odd L. The chain is the alternating word of LEN L+1, whose backward reading equals its NOT. At every LEN exactly one facing pair closes into a ring by itself, each member chaining into the other: the alternating pair. It is the trace of NOT applied at every tick, the one-bit twisted clock τ_1. The two constant words close alone, each chaining into itself.

Proof. For a palindrome x, OUT(x) = J(IN x), while IN(C x) = C(IN x). The chain therefore needs J(IN x) = C(IN x), so IN(x) reads backward as its own NOT. That forces IN(x) to have even LEN and, with x a palindrome, forces every neighbouring pair of bits to differ. A direction-faced pair closes in both orders only when IN(x) and OUT(x) both read the same both ways, which forces x to repeat with period two; the non-constant such words are the alternating ones. ■ **STANDS**; **RUNS** LEN 1–14.

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Table 2. Ordered chains of a contact with its own partner, into LEN L+1.

LEN L	from direction pairs (= non-constant palindromes at L+1)	from NOT pairs	self-closing pairs
1	0	2	1
2	2	0	1
3	2	2	1
4	6	0	1
5	6	2	1
6	14	0	1
7	14	2	1
8	30	0	1
9	30	2	1
10	62	0	1
11	62	2	1
12	126	0	1
13	126	2	1
14	254	0	1

6.3 Whole versus parts

Theorem 6.3 was first derived in an independent GPT audit (5 November 2026) in the juxtaposition model. It is re-derived and re-checked here, and Theorem 6.4 states the same law in the offset model.

Theorem 6.3 (§181 model). Let W = a|b|b|c with a, b, c of LEN n, $N = 2^n$, $r = 2^{\lceil n/2 \rceil}$, U = a|b and V = b|c.

- (i) The whole partner $P_{4n}(W)$ is J(c)|J(b)|J(b)|J(a) when s(W) = 0 and C(a)|C(b)|C(b)|C(c) when s(W) = 1. Both keep the middle faces equal, so facing the whole preserves composability.
- (ii) The whole selector is s(W) = [a = J(c)] ∧ [b = J(b)]. It is collective: it holds on rN paths, and on r(N−1) of them neither part reads the same both ways.
- (iii) Facing the parts separately and reversing their order, P(V)|P(U), keeps the middle faces equal if and only if s(U) = s(V). It fails on exactly 2N(N−1) paths.
- (iv) The partwise result differs from the whole partner on 2N(N−1) + r(N−1) + (N − r) paths.

Proof. (i) and (ii) follow from J(W) = J(c)|J(b)|J(b)|J(a) by comparing faces. (iii) With both parts direction-faced, the inner faces are J(b) and J(b). With both NOT-faced, a = J(b) and c = J(b) force a = c, and the inner faces are C(c) = C(a). With one of each, the inner faces are a face and its NOT. (iv) The three disjoint cases are: exactly

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one part NOT-faced, $2N(N-1)$; neither part NOT-faced while the whole is, $r(N-1)$; both parts NOT-faced while the whole is not, $N - r$. ■ **STANDS**; **RUNS** $n = 2–6$ (Table 3).

Table 3. Whole versus parts in the §181 model (every composable path).

n	paths N^3	whole reads same both ways (rN)	of which no part does ($r(N-1)$)	parts fail to compose ($2N(N-1)$)	whole ≠ parts
2	64	8	6	24	32
3	512	32	28	112	144
4	4,096	64	60	480	552
5	32,768	256	248	1,984	2,256
6	262,144	512	504	8,064	8,624

Theorem 6.4 (offset model). Let c be a contact of LEN L+1 with parts x = IN(c) and y = OUT(c). Facing the parts separately and reversing their order, P(y) then P(x), chains if and only if x and y have the same facing type. It fails on $4r_L - 2b_L$ contacts, where $r_L = 2^{\lceil L/2 \rceil}$, $b_L = 4$ for odd L and $b_L = 2$ for even L (b_L counts the contacts whose two parts both read the same both ways). The whole reads the same both ways while neither part does on $2^{\lceil (L+1)/2 \rceil} - 2$ contacts, which are exactly the chains of Theorem 6.1. **STANDS**; **RUNS** $L = 1–14$ (Table 4).

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Table 4. Whole versus parts in the offset model (every contact). The last column equals $(4r_L - 2b_L) + (2^{\lceil (L+1)/2 \rceil} - 2) + 2 \cdot [L \text{ odd}]$.

L	contacts	parts fail to chain	whole reads same both ways, parts do not	whole ≠ parts
1	4	0	0	2
2	8	4	2	6
3	16	8	2	12
4	32	12	6	18
5	64	24	6	32
6	128	28	14	42
7	256	56	14	72
8	512	60	30	90
9	1,024	120	30	152
10	2,048	124	62	186
11	4,096	248	62	312
12	8,192	252	126	378
13	16,384	504	126	632
14	32,768	508	254	762

A composed whole has a facing condition at its own LEN, a conjunction over all its mirror pairs; its parts carry only their own. The collective seam, a whole that reads the same both ways built from parts that do not, is in both models a contact composed with its own backward reading. In the §181 table, b = J(b) and c = J(a) make V = J(U). Theorem 6.1 and Theorem 6.3(ii) are the same path at two locations, the offset wheel and the transition table.

7 Count, offset and closure

Theorem 7.1 (the elementary step). Facing is a bijection of X_n onto itself, so it never changes the number of contacts. The offset is the only move between LENs: IN and OUT each map X_{n+1} two-to-one onto X_n, and extension gives every face of X_n two contacts in X_{n+1}. A change in the number of distinguishable contacts is therefore a change of LEN, and its elementary unit is a factor of two, one bit (ln 2 in natural units). No move in the ring multiplies the count by three. **STANDS**; **RUNS** LEN 1–10.

This answers the question the external audit put to the horizon derivation (Technical Note 1). Flipping an existing distinction (facing) and adding an independent one (offset) are different operations, and only the offset changes the count. A horizon area that counts the contacts of a binary register changes by whole bits.

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Theorem 7.2 (closure). A ring of k contacts has one independent closed route; the same contacts opened into a chain have none. The wheel B(n) has $2^{n+1} - 2^n + 1 = 2^n + 1$. **Proof.** The cycle rank of a connected graph is E – V + 1. ■ **STANDS**.

The closure is collective: no single contact owns it. Reversal maps every route of B(n) to a route run backward, so closed routes come in reversal pairs. Facing agrees with reversal except on contacts that read the same both ways, where Theorem 6.4 shows that part-by-part facing breaks. The facing of a route is therefore taken whole, at the route's own LEN. Sorting the $2^n + 1$ closed routes by how facing acts on them is the first operation of Section 14.

8 The center

8.1 The center of a word

Theorem 8.1. At odd LEN n = 2k+1, reversal fixes exactly one position, the center k, and every other position has a mirror partner. The center has none at its own LEN: (a) the palindromes split evenly on its value, 2^k with center 0 and 2^k with center 1; (b) the selector s_n does not contain it, so it never decides which reading faces a word; (c) as a one-bit contact it reads the same both ways, so its other face is its NOT. At even LEN no position is its own mirror, the center is the seam between positions k–1 and k, and nothing occupies it. **STANDS**; **RUNS** LEN 1–15.

The center is therefore a zero or a one, either value admissible, at odd LEN, and nothing at even LEN. The two alternate as LEN advances.

8.2 One offset later

Theorem 8.2. Grow an odd word of LEN 2k+1 by one bit. Grown forward, with a bit appended on the right, the old center at position k pairs with position k+1, the bit after it. Grown backward, with a bit prepended on the left, the old center moves to position k+1 and pairs with position k, the bit that preceded it. **STANDS** (index arithmetic).

At its own LEN the center has no reflection, and its other face is its difference, the NOT. Its partner arrives with the next offset as a neighbor: the bit after it or the bit before it, chosen by the direction of reading.

8.3 The center of the ring

Theorem 8.3. $N = 2^n$ is even, so no address is its own mirror, and the ring's center is the seam between addresses m–1 and m. The seam holds $r\star = 1^{\lfloor n/2 \rfloor - 1}\,0\,1^{\lceil n/2 \rceil}$, the all-ones word with a single 0 just left of the word's center, and its reversal: $R_n[m-1] = J(r\star)$ and $R_n[m] = r\star$. The edge across the seam, $J(r\star)|r\star$, reads the same both ways at LEN 2n, so in the §181 transition table its partner is its NOT, not its reflection. Of the $2 \times 2^{\lceil n/2 \rceil}$ native successor edges whose facing is not a successor edge, it is the only one that touches no palindrome of LEN n.

Proof. r_{m-1} is the largest canonical representative. A NOT-faced pair's representative is the complement of a palindrome that begins with 1, so it begins with 0 and lies below $r\star$ for $n \geq 3$. For a direction-faced pair, the representative is whichever of the word and its reversal has its first 0 further left. Every word whose zeros all lie at or right of the center either reads the same both ways or has a reversal with a 0 further left. Every

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direction-faced representative therefore has its first o at position ⌊n/2⌋−1 or earlier, and the largest such word is r⋆. Since r⋆ does not read the same both ways, P(r⋆) = J(r⋆). Read backward, J(r⋆)|r⋆ is J(r⋆)|J(J(r⋆)) = J(r⋆)|r⋆. ■ **STANDS**; **RUNS** n = 2–16 for the seam and n = 2–12 for the successor exceptions (the §181 count, re-verified).

At the center of the ring a word sits beside its own backward reading. Its single o lies on one side of the word's center in one reading and on the other side in the other reading. The edge joining them has no reflection of its own, and its other face is its difference.

8.4 The center of the paired register

Theorem 8.4. In the paired register of Section 9 at odd LEN, the center qubit is in the state $(|0\rangle + |1\rangle)/\sqrt{2}$ and is unentangled with the rest. Read by value it returns 0 or 1 with equal weight; read by NOT it returns the same answer every time (the expectation of NOT is 1). At even LEN there is no center qubit: the innermost pair together is pure, and each of its two qubits alone carries one bit. **RUNS**: at n = 13 the center's reduced state is [[½, ½], [½, ½]], with purity 1 and entropy 0; at n = 12 the innermost qubit alone carries 1.000000 bit and the innermost pair together 0.

8.5 Status of the center

Theorems 8.1–8.3: **STANDS**. Theorem 8.4: **RUNS**. **PULL**: the center is the one place in the paired register where the value reading is undetermined and the NOT reading is definite. The proposal under test is that this is where superposition enters: a position with no reflection at its own LEN carries its difference, not a value. Section 14, item 4 defines the next test.

9 The paired register and its readout curve

Definition. For LEN n, $|\Psi_n\rangle$ is the equal superposition of all palindromes of LEN n. At even n = 2k it equals $2^{-k/2}$ $\Sigma\, |x\rangle|J\, x\rangle$, summed over $x \in X_k$: every half-word beside its own backward reading, with bit i paired to bit n−1−i. At odd n the center qubit is $(|0\rangle + |1\rangle)/\sqrt{2}$.

Theorem 9.1. Read $|\Psi_n\rangle$ out in order from position 0. After k bits, the part read out and the remainder share exactly min(k, n−k) bits of entanglement. **Proof.** Each mirror pair is a maximally entangled pair that contributes one bit across any cut separating it. The cut after k bits separates the pairs (i, n−1−i) with i < min(k, n−k), and the odd center is unentangled. ■ **STANDS**; **RUNS** n = 12 and 13 (Table 5, Figure 1).

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Table 5. Bits shared between the part read out (first k bits) and the remainder. "Typical" is Page's average over random pure states of the same size.

k	paired register, n = 12	typical, n = 12	paired register, n = 13	typical, n = 13
0	0	0.00	0	0.00
1	1	1.00	1	1.00
2	2	2.00	2	2.00
3	3	2.99	3	2.99
4	4	3.96	4	3.98
5	5	4.82	5	4.91
6	6	5.28	6	5.64
7	5	4.82	6	5.64
8	4	3.96	5	4.91
9	3	2.99	4	3.98
10	2	2.00	3	2.99
11	1	1.00	2	2.00
12	0	0.00	1	1.00
13	—	—	0	0.00

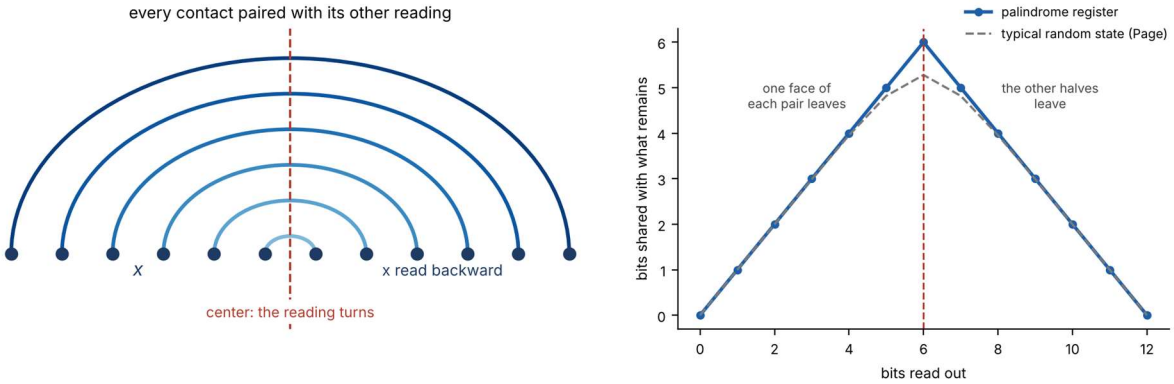


Figure 1. Left: every contact paired with its other reading (bit i with bit n−1−i), n = 12. Right: reading the register out in order; shared bits climb to the center and fall back to zero, on the upper envelope of the typical curve.

Up to the center, every bit that leaves takes one face of a pair and leaves the other behind, so the shared bits climb by one. After the center, every bit that leaves is the other half of one already gone, so they fall by one.

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The last bit out completes the first pair. A typical random state of the same size falls 0.72 bits short at the center; the paired register sits on the upper envelope.

9.1 The same path at other locations

Page's curve. Page (1993) showed that if a black hole evaporates without losing information, the entanglement between its radiation and its remainder climbs until about half the hole's entropy has left and then falls to zero. Theorem 9.1 is that curve exactly, with the turn at the center.

Hawking partners. Each outgoing Hawking quantum is paired with a partner at the reflected position across the horizon. Bisognano and Wichmann proved that the vacuum pairs the two sides of an acceleration horizon by a reflection. Israel's thermofield double for the eternal black hole pairs each state on one side with its reflected copy on the other, with thermal weights where $|\Psi_n\rangle$ has equal ones.

The funnel. A constant-time slice of the full black-hole geometry is Flamm's funnel doubled: two funnels joined at a throat, the Einstein–Rosen bridge. The throat is the ring the reflection holds fixed. In this map the center the pairing turns around is the throat, the horizon itself, and not the singularity, which does not lie on that slice.

10 Landing: the NOT at the Planck floor and the horizon register

10.1 The floor

A distinction at separation x needs energy $E \geq \hbar c/x$ to be read, and it stays readable only while $x \geq 2GE/c^4$; with more energy than that inside x, the distinction hides behind its own horizon. The first bound rises as x shrinks and the second falls, and they cross at $x_\star = \sqrt{(2\hbar G/c^3)} = \sqrt{2}\,\ell_P$. The contract fixes the crossing; the factor $\sqrt{2}$ comes from the particular estimates. Mead (1964) derived the same limit as a bound on measuring position.

10.2 Count, not curvature

A horizon sits at the floor in count, not in curvature. The Kretschmann curvature at a Schwarzschild horizon is $12/r_s^4$, small for a stellar-mass hole. What the horizon saturates is the number of distinctions its area allows, $S = A/(4\ell_P^2)$ in natural units.

10.3 The elementary step

By Theorem 7.1, a change in the horizon's count is a change of LEN, and its elementary unit is one bit. With $S = A/(4\ell_P^2)$ this gives $\Delta A = 4 \ln 2\ \ell_P^2$ ($\alpha = 4 \ln 2 \approx 2.7726$), with 2^N states at level N. With the first law $dM = (\kappa/8\pi)\, dA + \Omega_H\, dJ$, a Kerr remnant trades $l = m = 2$ quanta only at

$$\omega_N = \kappa\, \alpha\, N / (8\pi) + 2\Omega_H \quad (G = c = 1,\ \hbar = \ell_P^2,\ N \text{ a whole number}).$$

10.4 The seal

predictions13 was sealed on 5 November 2026 at 05:32:22 UTC, before any search for horizon-spectrum or Planck-grid data (SHA-256 in Appendix A). P13.1: $\alpha = 4 \ln 2$. P13.2: the line formula above. P13.3: no published ringdown analysis excludes $\alpha = 4 \ln 2$. P13.4: no Planck-grid signature has been reported. P13.4 was known to the sealer beforehand and protects less than P13.3.

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10.5 The check

GW250114: remnant spin $\chi_f = 0.68 \pm 0.01$ (90%), and the 220 tone measured 0.02 ± 0.02 above the Kerr value for that remnant (90%), about 2% precision (LVK 2025). Kerr tones were taken exact from the qnm package; at $\chi = 0.68$, $M\omega_{220} = 0.523975$. Mass cancels, so the test compares the offset of each line, $\delta_{req}(N) = \omega_N/\omega_{220} - 1$, with the measured offset.

Table 6. Horizon-area combs against the GW250114 220 tone ($\sigma = 0.02/1.645$).

step α	line spacing (share of tone)	nearest line N	offset of that line	distance from measured tone	spins with a line in the 90% band
4 ln 2 (binary, sealed)	4.45%	6	+1.60%	0.33σ	all
4 ln 3 (Hod)	7.06%	4	+3.11%	0.91σ	all
8π (Maggiore)	40.4%	1	+15.2%	10.9σ	none
15.6 (Laghi et al. fit)	25.1%	1	−0.07%	1.70σ	43%
Kerr, no quantization	—	—	0	1.65σ	—

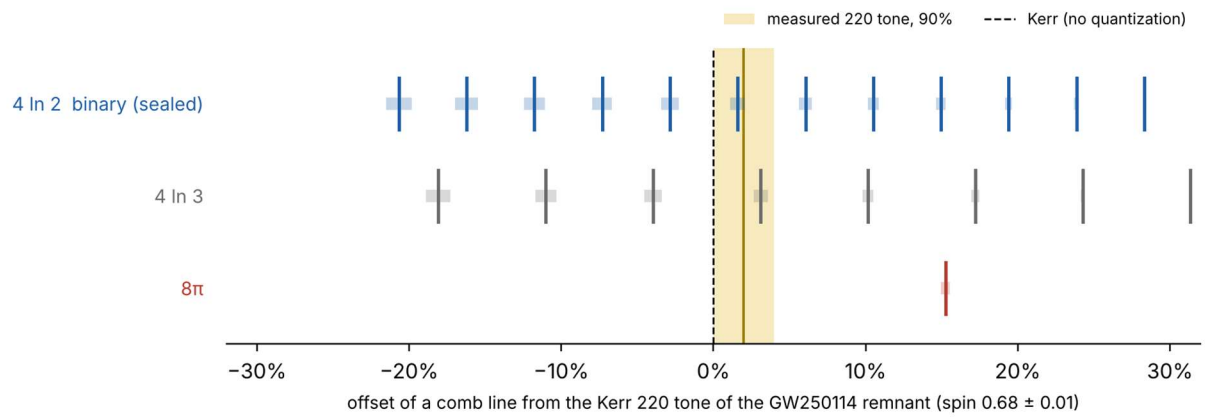


Figure 2. The measured 220 tone (gold band, 90%) against three horizon-area combs. Each tick is a frequency the horizon is allowed to ring at; pale bars show how the ticks move across the spin uncertainty.

Published analyses: Laghi et al. (2021), using the events to 1 October 2019, found α = 15.6 (+20.5/−13.3) with log odds 0.1 ± 0.6, which is not informative. The LVK GW250114 spectroscopy paper does not test quantization. Planck-grid searches: LHAASO's GRB 221009A gives $E_{QG,1} > 10\ E_{Pl}$ and $E_{QG,2} > 6 \times 10^{-8}\ E_{Pl}$ (95%), and the Fermilab Holometer excluded Hogan's pixel-noise model.

10.6 Status

P13.3 **STANDS**. The comb check **RUNS**: the binary comb survives, and at 2% precision it cannot fail, because its lines are 4.45% apart; this is survival, not support. The 8π step fails in the Foit–Kleban emission-line model. The fitted 15.6 lands **ELSEWHERE**: it is the α whose first line sits on the Kerr tone of a spin-0.68 remnant, and most detected remnants spin near 0.7. P13.4 **STANDS** as a retrodiction. **OPEN**: binary against ternary needs

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the 220 tone to well under 1% on one remnant, or a population of remnants with spread spins, since each spin slides the comb against the Kerr tone.

11 The interface catalog and the motion reader

11.1 The affine catalog

Two transformations of LEN-L words implement the same interface when a relabeling that respects XOR carries one onto the other, that is when they are conjugate in the affine group AGL(L,2). The interfaces of LEN L are its conjugacy classes. Table 7 lists them for LEN 1–3. **RUNS**: every affine map enumerated, 2 at LEN 1, 24 at LEN 2 and 1,344 at LEN 3.

Table 7. Affine interfaces by LEN (cycle type on the 2^L words, number of maps, order, fixed words).

LEN	#	cycles	maps	order	fixed	placement
1	0	1,1	1	1	2	identity
1	1	2	1	2	0	NOT; P_1
2	0	1,1,1,1	1	1	4	identity
2	1	2,1,1	6	2	2	
2	2	2,2	3	2	0	translations; $P_2 = C_2$
2	3	3,1	8	3	1	
2	4	4	6	4	0	
3	0	1^8	1	1	8	identity
3	1	2,2,1,1,1,1	42	2	4	mate map Φ_A of the 17-bit sector
3	2	2,2,2,2	7	2	0	translations; C_3
3	3	2,2,2,2	42	2	0	facing P_3; the sector partner; the J-pair partner
3	4	3,3,1,1	224	3	2	
3	5	4,2,1,1	168	4	2	
3	6	4,4	84	4	0	
3	7	4,4	168	4	0	
3	8	6,2	224	6	0	
3	9	7,1	192	7	1	mirror twin of #10 (x^3+x+1)
3	10	7,1	192	7	1	mirror twin of #9 (x^3+x^2+1)

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Nine of the eleven LEN-3 interfaces contain their own inverses. The two order-7 interfaces are each other's inverse: their way home is a different interface.

Theorem 11.1 (mirror twins at every LEN from 3). For every $L \geq 3$ the catalog contains an interface whose way home is a different interface. **Proof.** Multiplication by a primitive element α of $GF(2^L)$ is a linear map of order $2^L - 1$, a Singer cycle, and its inverse is multiplication by α^{-1}. Conjugate affine maps have conjugate linear parts, and conjugate linear maps have equal characteristic polynomials. α^{-1} has the minimal polynomial of α only when $\alpha^{-1} = \alpha^q$ with $q = 2^j$ for some j, that is when $2^j \equiv -1 \pmod{2^L - 1}$. For $L \geq 3$ every 2^j with $0 \leq j < L$ is smaller than $2^L - 2$, so no such j exists. At $L = 2$, $2 \equiv -1 \pmod 3$ and the inverse is a conjugate. ■ **STANDS**.

11.2 The motion reader (predictions12)

Procedure P8 was tested under seal (predictions12, Appendix A) on six machines over 2^{18} states each: a reader, a carry and four paired cells. A1 is the ring in 3-bit codes, A2 the ring built with a scrambled operation table, and A3 A1's run written in 4-bit words. B is a J-pair partner rule, C a random reversible map, and D B's outputs remapped to copy A1's value histogram. The value reader compares histograms of values read. The motion reader compares revisit lags read only as same or different, orbit lengths, and places used.

Table 8. Distance to A1 by values and by motion.

machine	what it is	value distance	motion distance
A2	the ring, scrambled operation table	0.026	0
A3	the ring, 4-bit words	0.504	0
B	J-pair partner rule (the ring relabeled)	0.033	0.024 (start noise)
C	random reversible map	0.022	0.098
D	B dressed in A1's value histogram	0.011	0.024 (start noise)

- W1 **RUNS**: the three ring implementations differ by values and agree by motion, with identical orbit lengths.
- W2: the random map C separates by motion (10 of its 17 orbits are odd, against 0 of 5,624 for the ring). B lands **ELSEWHERE**: it is the ring under other labels. Its orbit-length spectrum equals the ring's, and four XOR-respecting relabelings carry the ring's partner and mate maps exactly onto B's.
- W3: the value reader ranks D (0.011) and the random map C (0.022) closer to the ring than the ring's own second program A2 (0.026).
- W4 **RUNS**: on all 128 closed walks of A1, the average return lag equals the number of places the walk uses, exactly (Kac).
- W5 **ELSEWHERE**: a value-pair distance (0.080 between A1 and B) separates two implementations of one interface.
- Every ring orbit is even. The ring obeys the contract of a closed chain of NOTs, $N^m = I$ only for even m; the random map does not.

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12 Technical notes

Technical Note 1 (independent GPT audit of predictions13, 5 November 2026)

Objects: the contract statement "every patch of a horizon sits at the Planck floor"; the factor $\sqrt{2}$ in $x^\star$; the step from one NOT to one bit of area; the use of the Foit–Kleban line identification; the selection of the nearest comb line.

Disposition: the statement is corrected to count, not curvature (Section 10.2). The factor is attributed to the estimates (10.1). The step is supplied by Theorem 7.1: facing flips and keeps the count, while the offset adds an independent bit. The 8π result is stated inside the Foit–Kleban model, and the binary comb's proximity is reported as survival, not support (10.6).

Technical Note 2 (Hod 1998; Motl 2002)

Object: the binary step $\alpha = 4 \ln 2$. The highly damped Schwarzschild tones approach $T_H \ln 3$ exactly, which Hod read as $\alpha = 4 \ln 3$. This concerns the asymptotic tones, not the fundamental 220 tone used in Section 10.5, and it is the strongest standing objection to the binary step.

Disposition: binary against ternary is OPEN. In the ring a factor of three has no move (Theorem 7.1), so this test reads directly on the claim that difference forces binary.

Technical Note 3 (equal weights)

Object: the comparison in Section 9.1. $|\Psi_n\rangle$ has equal weights, while the thermofield double carries thermal weights on the same pairing. The comparison is to the pairing and to the shape of the readout curve.

Technical Note 4 (ranges)

Object: every RUNS label. Each is exhaustive over its stated range. Statements claimed for every LEN carry proofs and the label STANDS.

13 Ledger

Table 9. Status of every result in this manual.

ID	Result	Status
F1	Facing exchanges interface and implementation for direction-faced contacts; NOT-faced contacts have both faces negated (Thm 5.1)	STANDS; RUNS LEN 2–14
F2	NOT faces a share $2^{\lfloor n/2 \rfloor - n}$ of contacts (Thm 5.2)	STANDS; RUNS LEN 1–14
F3	Facing is a pure NOT at LEN 1–2, affine at LEN 3 (catalog #3), degree $\lfloor n/2 \rfloor$ from LEN 4; the center never enters the selector (Thm 5.3)	STANDS; RUNS LEN 1–12
F4	A contact chained with its own backward reading through a palindromic hinge reads the same both ways; one-to-one with non-constant palindromes one level up (Thm 6.1)	STANDS; RUNS LEN 1–14

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ID	Result	Status
F5	NOT pairs chain only on the alternating word; the alternating pair is the only self-closing facing pair (Thm 6.2)	STANDS; RUNS LEN 1–14
G1	Whole facing preserves composability; part-by-part facing composes iff facing types agree; counts in the §181 model (Thm 6.3; GPT audit, re-verified)	STANDS; RUNS n = 2–6
G2	The same law in the offset model, with counts (Thm 6.4)	STANDS; RUNS L = 1–14
K1	Facing keeps the count; the offset changes it by one bit per LEN step; no move multiplies by three (Thm 7.1)	STANDS; RUNS LEN 1–10
K2	Cycle rank: ring 1, chain 0, wheel $2^n + 1$ (Thm 7.2)	STANDS
C1	Word center: free bit at odd LEN, outside the selector, faced by NOT; a seam at even LEN (Thm 8.1)	STANDS; RUNS LEN 1–15
C2	One offset later the center pairs with a neighbor chosen by reading direction (Thm 8.2)	STANDS
C3	Ring center: the seam holds J(r⋆) | r⋆; the edge across reads the same both ways and is faced by NOT; the only successor exception touching no palindrome (Thm 8.3)	STANDS; RUNS n = 2–16
C4	Register center: equal mix, unentangled, definite under NOT (Thm 8.4)	RUNS n = 12, 13
R1	The paired register reads out along min(k, n−k) (Thm 9.1)	STANDS; RUNS n = 12, 13
A1	Affine catalog: 2, 5, 11 interfaces at LEN 1–3; P_3 is #3	RUNS
A2	Interfaces whose way home is a different interface exist at every LEN from 3 (Thm 11.1)	STANDS
M1	Motion identifies interfaces and values do not (predictions12)	RUNS; B ELSEWHERE
H1	No published ringdown analysis excludes $\alpha = 4 \ln 2$ (P13.3)	STANDS
H2	GW250114 combs: binary 0.33σ, $4 \ln 3$ 0.91σ, 8π 10.9σ in the Foit–Kleban model	RUNS
H3	The fitted $\alpha = 15.6$ is the first-line alignment for spin 0.68	ELSEWHERE
H4	No Planck-grid signature reported (P13.4)	STANDS (retrodiction)
O1	Binary against ternary at horizons	OPEN
O2	The center as the place where superposition enters	PULL
O3	Facing split of the wheel's $2^n + 1$ closed routes	PULL
O4	LEN-4 affine catalog (322,560 transformations)	OPEN

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14 How it continues

1. **Split the closed routes.** The wheel B(n) has $2^n + 1$ independent closed routes. Reverse every route, face it whole at its own LEN, and sort the route space by how facing acts on it. This extends Theorems 6.1–6.4 from single contacts to routes.
2. **Follow the collective selector upward.** A composed whole's selector is a conjunction over its mirror pairs (Theorem 6.3(ii)). Map how the selectors of k composed contacts combine and where the collective seams sit at each LEN.
3. **Complete the LEN-4 catalog.** Sort the 322,560 affine transformations into interfaces, and locate where facing leaves the affine catalog (P_4 has degree 2).
4. **Test the center.** The two NOT-definite centers, $(|0\rangle + |1\rangle)/\sqrt{2}$ and $(|0\rangle - |1\rangle)/\sqrt{2}$, give registers that no value reading of the center distinguishes and that a NOT reading always distinguishes. Determine which ring motions carry that difference and which erase it: the native successor T_n, the twisted clock τ_n, and facing P_n applied to the register. Seal before running.
5. **Binary against ternary at horizons.** Seal the comb test before each new loud ringdown. One remnant with the 220 tone to well under 1%, or a population with spread spins, decides between $4 \ln 2$ and $4 \ln 3$.
6. **The boundary-driven compiler.** Implement the three operations as the compiler's verbs: bind (do two contacts meet, P3), compose (what the connection does, P4 and Section 6) and close (what new collective routes exist, Section 7). A completed operation exposes its boundary as the next requirement.

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Appendix A Seals

Sealed predictions referenced in this manual.

Seal	SHA-256	Sealed
predictions12	9e7a7300ad4ce33a5f10a76bca183a3fc041481a1a8bf39db42802d92e6d0ed6	5 November 2026
predictions13	52e37183273b9977ed7a968a96f80a5a5b0d05e127e25a025af5b299d602e64b	5 November 2026, 05:32:22 UTC

The sealed text of predictions13 follows. The hash covers the exact bytes of the file predictions13.txt, which is kept with the project record.

```
predictions13: NOT at the Planck floor, landed on a horizon
Sealed 2026-11-05, before any search for horizon-spectrum or Planck-grid data.


Contract side
A distinction has to be paid for, and the payment rises as the distinction gets finer.
The payment bends the field that has to read the distinction. The two gradients run in
opposite directions and touch at one scale: the floor. A horizon is a closed surface on
which every patch sits at that floor, so its content is pure interface. The smallest
change a horizon can make is one NOT: one bit, ln 2 nats.


Exchange rates taken from the field (comparators, not derived here):
S = A / (4 l_P^2) nats, and the first law dM = (kappa / 8 pi) dA + Omega_H dJ.


P13.1  Horizon area moves in equal steps dA = 4 ln2 l_P^2, so alpha = 4 ln 2 = 2.7726,
    with 2^N states at level N.
P13.2  A Kerr remnant then trades l = m = 2 quanta only at
    omega_N = kappa * alpha * N / (8 pi) + 2 Omega_H   (G = c = 1, hbar = l_P^2), N whole.
P13.3  Landing. Published LIGO-Virgo-KAGRA analyses that test horizon-area quantization
    against ringdown data do not exclude alpha = 4 ln 2. Fails if any such analysis
    reports alpha = 4 ln 2 excluded, or strongly disfavoured against no quantization.
P13.4  The floor is a bound, not a grid, because no location carries a value. No experiment
    has reported a Planck-grid signature: the Fermilab Holometer saw no Planck-scale
    jitter, and gamma-ray-burst timing puts the first-order energy-dependent light-speed
    scale above the Planck energy. Fails if either reported a detection.
    (The sealer already knew this one from memory, so it protects less than P13.3.)
```

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Appendix B Reference implementation

The module below implements the operators, both contact models, the native ring and the center, and checks the main theorems exhaustively in small ranges. It uses only the Python standard library and runs in about half a minute. Expected output: ten lines marked PASS, then ALL CHECKS PASS.

```
#!/usr/bin/env python3
"""Facing, composition and the center in native binary rings: reference implementation.

Dean A. Kulik, QuHarmonics Research Group (ORCID 0009-0003-3128-8828), 5 November 2026.
Standard library only. A word is a tuple of bits, position 0 on the left.
Run:  python3 nexus_facing.py   (about half a minute; every check is exhaustive in its range)
"""
from itertools import product
from math import ceil



# ---------------------------------------------------------------------- types and operators
def words(n):
  return [tuple(w) for w in product((0, 1), repeat=n)]


def J(w):         return w[::-1]              # read backward (reversal)
def C(w):         return tuple(1 - b for b in w)     # NOT every bit (complement)
def is_pal(w):      return J(w) == w              # reads the same both ways
def P(w):         return C(w) if is_pal(w) else J(w)  # facing (native partner)
def facing_type(w):  return 'NOT' if is_pal(w) else 'direction'
def IN(e):        return e[:-1]               # offset model: face received
def OUT(e):       return e[1:]                # offset model: face handed on
def chains(e, f):   return OUT(e) == IN(f)
def chain(e, f):    return e + f[-1:]



# ---------------------------------------------------------------------- native ring
def native_ring(n):
  """Native Binary Rings, eqs. 2-4: representatives in order, seam, partners mirrored."""
  reps = sorted(w for w in words(n) if w < P(w))      # smaller member of each pair
  tail = [P(w) for w in reversed(reps[:-1])]
  return reps[:-1] + [P(reps[-1]), reps[-1]] + tail



# ---------------------------------------------------------------------- §181 model: a->b->c is a|b|b|c
def whole_face(a, b, c):
```

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```
    n = len(a)
    F = P(a + b + b + c)
    return F[:n], F[n:2 * n], F[2 * n:3 * n], F[3 * n:]


def parts_face(a, b, c):
    """Face each transition separately, then reverse their order."""
    n = len(a)
    PU, PV = P(a + b), P(b + c)
    return PV[:n], PV[n:], PU[:n], PU[n:]



# ---------------------------------------------------------------- the center
def word_center(n):
    """Odd LEN: the middle position. Even LEN: None (a seam between n/2-1 and n/2)."""
    return (n - 1) // 2 if n % 2 else None


def ring_seam(n):
    R = native_ring(n)
    m = len(R) // 2
    return R[m - 1], R[m]



# ---------------------------------------------------------------- self-test
def check(label, ok):
    print(f'  {"PASS" if ok else "FAIL"}  {label}')
    assert ok


def test_facing():
    check('facing is a fixed-point-free involution, LEN 1-12',
          all(P(P(w)) == w and P(w) != w for n in range(1, 13) for w in words(n)))
    ok = True
    for n in range(2, 13):
        R = native_ring(n)
        ok &= all(R[2 ** n - 1 - a] == P(R[a]) for a in range(2 ** n))
    check('native ring address law R[N-1-a] = P(R[a]), n = 2-12', ok)
    ok = True
    for n in range(1, 12):
        for e in words(n + 1):
            p = P(e)
            if is_pal(e):
                ok &= (IN(p), OUT(p)) == (C(IN(e)), C(OUT(e)))
            else:
```

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```
            ok &= (IN(p), OUT(p)) == (J(OUT(e)), J(IN(e)))
    check('exchange: direction-faced partner has IN = J(OUT), OUT = J(IN); '
        'NOT-faced negates both, LEN 2-12', ok)
    check('NOT faces a share 2^(ceil(n/2)-n) of contacts, LEN 1-12',
        all(sum(map(is_pal, words(n))) == 2 ** ceil(n / 2) for n in range(1, 13)))


def test_composition():
    ok = True
    for L in range(1, 12):
        pairs = {frozenset((w, P(w))) for w in words(L)}
        dch, nch, closers = [], [], 0
        for pr in pairs:
            a, b = sorted(pr)
            hits = [chain(x, y) for x, y in ((a, b), (b, a)) if chains(x, y)]
            (nch if is_pal(a) else dch).extend(hits)
            closers += len(hits) == 2
        const = {(0,) * (L + 1), (1,) * (L + 1)}
        nonconst_pal = {w for w in words(L + 1) if is_pal(w)} - const
        alt = {tuple((i + s) % 2 for i in range(L + 1)) for s in (0, 1)}
        ok &= sorted(dch) == sorted(nonconst_pal)
        ok &= set(nch) == (alt if L % 2 else set()) and closers == 1
    check('a contact chained with its own backward reading reads the same both ways; '
        'one-to-one with non-constant palindromes one level up; NOT pairs chain only '
        'on the alternating word; one self-closing pair, LEN 1-11', ok)
    ok = True
    for n in range(2, 5):
        N, r = 2 ** n, 2 ** ceil(n / 2)
        seams = breaks = lone = differ = 0
        for a, b, c in product(words(n), repeat=3):
            wf, pf = whole_face(a, b, c), parts_face(a, b, c)
            sW, sU, sV = is_pal(a + b + b + c), is_pal(a + b), is_pal(b + c)
            ok &= wf[1] == wf[2] and sW == (a == J(c) and b == J(b))
            ok &= (pf[1] == pf[2]) == (sU == sV)
            seams += sW
            breaks += pf[1] != pf[2]
            lone += sW and not sU and not sV
            differ += pf != wf
        expected = (r * N, 2 * N * (N - 1), r * (N - 1), 2 * N * (N - 1) + r * (N - 1) + N - r)
        ok &= (seams, breaks, lone, differ) == expected
    check('whole versus parts (§181 model): whole facing keeps the middle matched; parts '
        'compose iff their facing types agree; counts rN, 2N(N-1), r(N-1), '
        '2N(N-1)+r(N-1)+N-r, n = 2-4', ok)
    ok = all(chains(P(OUT(c)), P(IN(c))) == (is_pal(IN(c)) == is_pal(OUT(c)))
             for L in range(2, 12) for c in words(L + 1))
```

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```
    check('whole versus parts (offset model): faced parts chain iff their facing types '
        'agree, LEN 2-11', ok)


def test_center():
    ok = True
    for n in range(1, 14, 2):
        k = word_center(n)
        pals = [w for w in words(n) if is_pal(w)]
        ok &= sum(w[k] for w in pals) == len(pals) // 2          # the center is free
        flip = lambda w: w[:k] + (1 - w[k],) + w[k + 1:]
        ok &= all(is_pal(w) == is_pal(flip(w)) for w in words(n))  # and outside the selector
    check('odd LEN: the center is free (0 or 1) and never enters the facing selector; '
        'even LEN: a seam, LEN 1-13', ok)
    ok = True
    for n in range(2, 13):
        left, right = ring_seam(n)
        rstar = (1,) * (n // 2 - 1) + (0,) + (1,) * (n - n // 2)
        edge = left + right
        ok &= right == rstar and left == J(rstar) and is_pal(edge) and P(edge) == C(edge)
    check('ring center: the seam holds J(r*) | r*, r* all ones with one 0 just left of '
        'center; the edge across it reads the same both ways and is faced by NOT, '
        'n = 2-12', ok)


def test_count():
    ok = True
    for n in range(1, 11):
        ok &= len({P(w) for w in words(n)}) == 2 ** n
        ok &= all(sum(1 for e in words(n + 1) if IN(e) == f) == 2 for f in words(n))
    check('facing keeps the count; the offset maps two contacts onto each face '
        '(one bit per LEN step), LEN 1-10', ok)


def main():
    print('Native binary ring facing: self-test')
    test_facing()
    test_composition()
    test_center()
    test_count()
    print('ALL CHECKS PASS')



if __name__ == '__main__':
    main()
```

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Appendix C Reproduction

- Environment: Python 3.11 or later. The reference implementation needs only the standard library. The register check (Theorems 8.4 and 9.1) uses NumPy. The horizon check (Section 10.5) uses NumPy and qnm 0.4.4 (Stein 2019) for exact Kerr tones.
- Core theorems: python3 nexus_facing.py (standard library only).
- Full audit with larger ranges (Sections 5–8), run from the folder holding the scripts: python3 audit.py (sections A–E, including the §181 counts to n = 6), python3 seam.py (ring center, n = 2–16), python3 degree.py (facing degree, LEN 1–12), python3 facing_map.py (Tables 1, 2 and 5; writes S12.npy). ringops.py holds the integer-encoded operators the audit scripts share.
- Horizon comb (Section 10.5): python3 comb.py. Inputs: $\chi_f = 0.68 \pm 0.01$ and $\delta\tilde{f}_{220} = 0.02 \pm 0.02$ (90%). Figures: python3 comb_fig.py (Figure 2) and python3 rainbow_fig.py (Figure 1, after facing_map.py).
- Record: the computations in this manual were run in a Claude (Anthropic) session on 5 November 2026. The whole-versus-parts derivation of Section 6.3 was supplied by an independent GPT audit and re-verified here.

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